# 75 Lies by Renee Hannon About Cohler & Soon (2026)

## Companion to the Forensic Audit (color-coded HTML).

*By Jonathan Cohler* • Nov 11, 2026

Renee Hannon has published two Substack posts (Nov 2 and Nov 4, 2026) attacking Cohler & Soon (2026), *Mathematical Falsification of the Fossil-Dominant Hypothesis for Rising Atmospheric CO2 Using the Keeling Relation*. Both posts misrepresent what the paper says, misrepresent what the cited primary sources show, and rest on a statistical construction (the "cumulative Keeling regression") that produces artifactual intercept drift from narrow 1/C windows and nothing else. This document catalogs every misrepresentation against the primary sources: Cohler and Soon (2026), Koutsoyiannis (2024), Böhm et al. (2002), Francey et al. (1999), and Köhler et al. (2006).

Hannon is a co-author of Engelbeen, Hannon & Burton (2024), a non-peer-reviewed CO2 Coalition report arguing that fossil-fuel combustion is responsible for nearly all ($f \approx 1$) of the atmospheric CO2 growth since the industrial revolution. This position is variously called *fossil dominance* or *fossil-fuel dominance* and is the IPCC-endorsed framework. Cohler and Soon (2026) falsifies it mathematically through the Keeling intercept. A companion Cohler and Soon paper audits EHB (2024) line by line and finds every one of its nine lines of evidence invalid. Hannon's two Substack posts attack Cohler and Soon (2026) and appear to be an effort to resurrect EHB's conclusions.

## On language: a Keeling regression is a computation, not a claim

A Keeling regression is a mathematical operation. It takes a paired dataset of $\delta^{13}C$ and 1/[CO2] values and returns two numbers: a slope and an intercept. There is no "claim" involved any more than there is a claim involved in reporting that 2 + 2 = 4. The arithmetic is done correctly or incorrectly; the result is what it is.

Hannon's posts repeatedly recast mathematical operations as "claims," "arguments," "assumptions," or things the authors "rely on." Phrases like "Cohler-Soon claim…," "their derivation assumes…," "the paper's premises," "their interpretation of the Keeling intercept" treat computations as opinions open for debate. They are not opinions. The results of a Keeling regression on a published dataset are facts. Anyone can run the same regression on the same data and get the same intercept. That is what it means for a thing to be a computation.

The only honest responses to a reported regression result are:

1. Accept the result.
2. Show the arithmetic was done wrong, step by step.
3. Show the input data is mis-measured or mis-identified, and argue what the correct data is.
4. Show that the reported result is not statistically significant: that the uncertainty on the fit swallows the quoted number, that the fit explains almost none of the variance in the data ($R^2$ near zero), or that the uncertainty range on the number includes the opposite conclusion. A regression result is never just a number. It is a number plus its uncertainty. Reporting the number without the uncertainty is reporting half a result, and treating the half result as a full one is a form of arithmetic error.

**Response (4) is the one that does most of the work in this dispute.** Cohler and Soon (2026) reports standard errors, $R^2$ values, confidence intervals, and statistical significance tests throughout. Every reported intercept carries its uncertainty. Every reported trend carries a null-distribution test. Hannon's posts do the opposite. The intercepts she reports from her cumulative Keeling regressions and from her 1965 split carry no uncertainty bars at all. Her own chart plots $R^2$ on the right axis as a near-silent admission that the early-

window intercepts have $R^2$ near zero (the fitted intercept is noise, not a measurement), but she does not report the standard errors that would make the significance failure explicit to the reader. When she does report a significance test (sup-F = 36.4, bootstrap p = 0.03 for her 1965 "break"), the test is applied to a setup that cannot carry the inference: her "break" coincides with the Law Dome data-source transition from ice-core bubbles to firn air to direct atmospheric sampling, not with a change in the atmospheric signature. Reporting a point estimate without its uncertainty, and running a significance test on a setup that cannot carry the inference, are both forms of arithmetic concealment. They prevent the reader from seeing that the number quoted is not a result.

"I disagree," "this is a claim," "this needs to be reconciled with other evidence," "this is an assumption that may be questioned" are responses to opinions, not computations. Converting a computation into an opinion by calling it a "claim" is the technique. It is the device by which someone who cannot show an arithmetic error nevertheless appears to object to the result. This technique is especially corrosive because it can be repeated indefinitely without engaging any arithmetic. Every time it is repeated, mathematical reality looks a little more like opinion in the reader's mind, and the next unsupported objection seems a little more legitimate. This is how 2 + 2 = 4 gets dragged into territory where it can be "disputed."

**This is a named and well-studied pattern of disinformation, not a quirk of one writer.** In the sociology of science, it is called *manufactured doubt*. The phrase comes from a 1969 internal memo at the Brown & Williamson tobacco company, which stated: *"Doubt is our product, since it is the best means of competing with the 'body of fact' that exists in the mind of the general public."* The playbook was refined through decades of corporate PR on health and environmental questions and is documented in David Michaels, *Doubt Is Their Product* (2008), and Naomi Oreskes and Erik Conway, *Merchants of Doubt* (2010). The academic field that studies deliberately induced ignorance is called *agnotology*, coined by Stanford historian Robert Proctor.

The same family of techniques is studied in military and state-level information-warfare research. The RAND Corporation's analysis of the Russian propaganda model, the *firehose of falsehood* (Paul and Matthews, 2016), and the NATO Strategic Communications Centre of Excellence publications on cognitive warfare document their systematic deployment by state actors. The lineage is continuous: the methods were developed in corporate PR, refined across industry-funded science denial from tobacco through acid rain, ozone, pesticides, and now climate attribution, and have since been adopted and extended by state information operations. So the techniques on display in these two Substack posts are not idiosyncratic. They are a specific instance of a well-mapped methodology.

The variant on display here is the one directed specifically at mathematics. Call it *epistemological downgrading*: the technique that converts a computation (whose result is determinate) into a "claim" (which may be disputed), so that someone who cannot find an arithmetic error can nevertheless appear to object. This is what pseudo-skeptics do when they cannot find an error. It is also the device by which 2 + 2 = 4 can be dragged into territory where it seems to be a matter of opinion. The reader who absorbs this one point will recognize the same technique wherever it recurs, in the climate-attribution dispute and well beyond it.

Throughout this Catalog, Cohler and Soon (2026)'s mathematical results are treated as the computations they are. Koutsoyiannis performed a Keeling regression on atmospheric data from Etheridge, Francey, and Keeling-Whorf, and obtained an intercept of −13.3‰. That is a fact. Köhler performed a Keeling regression on Law Dome LD ANT ice-core atmospheric data and obtained $y_0 = -13.1‰$ with $r^2 = 96\%$. That is a fact. The Scripps four-site paired measurements give −13.2 ± 0.2‰. That is a fact. The full-record Keeling regression on 101 annual Law Dome + CGAA + Cape Grim pairs over 1006–2025 gives −13.42 ± 0.07‰ with $r^2 = 0.988$. That is a fact. These are not claims. They are the outputs of defined operations applied to published datasets.

The two-component mass partition equation (fossil fraction f solved from the measured intercept and the standard $\delta^{13}C$ values for fossil and ocean-outgassed CO2) is likewise not an opinion. It is a consequence of

mass conservation and the definition of a flux-weighted mean. The numerical result follows by substitution. The only way to object to it is to reject mass conservation, reject the standard $\delta^{13}C$ values from Mook 1986 and Zhang 1995 and the rest of six decades of carbon-cycle literature, or reject the published atmospheric datasets. Hannon rejects none of these. She objects to the result anyway. That is what makes it dishonest rather than merely mistaken.

Where the forensic notes below use the word "claim," it refers only to Hannon's own propositional statements about physical attribution, which are claims because they are not computations. Cohler and Soon (2026)'s mathematical outputs are not claims. They are results.

## Executive summary

Four load-bearing devices carry Hannon's two posts. All four fail against the primary sources.

1. **The central strawman.** Hannon writes as if Cohler and Soon (2026) performs a Keeling regression on sclerosponge measurements. A Keeling regression is a mathematical operation on paired atmospheric $\delta^{13}C$ and 1/CO2 data; it is not performed on seawater DIC data by anyone in the literature, because DIC is not atmospheric CO2. The paper's atmospheric Keeling regression is Koutsoyiannis's, computed on real atmospheric data from Antarctic ice core, firn air, and direct air measurements (Etheridge 1996, Francey 1999, Keeling-Whorf 2001). The sclerosponge data record surface-water DIC and provide a five-century stability check through the standard gas-DIC fractionation of approximately 9‰. Cohler and Soon (2026) states this explicitly in its Section 4.2. Hannon spends two sections of her second post refuting a position no one in this literature holds.
2. **The statistical fraud: cumulative Keeling regression.** Her "cumulative Keeling regression" takes early windows where atmospheric CO2 barely moved. In those windows the 1/C variable spans essentially no range, so the regression's extrapolated intercept is statistically meaningless. As the window widens and the 1/C range grows, the regression stabilizes toward its true value. The "drift" from −11‰ to −13‰ she displays is the regression stabilizing. Not a physical shift in the atmospheric signature. Her own $R^2$ curves show this directly: $R^2$ is low in the early windows and rises as the window widens.
3. **The Köhler 2006 omission.** Köhler et al. (2006) performed a Keeling regression directly on Law Dome LD ANT ice-core data and reported $y_0 = -13.1$‰ with $r^2 = 96\%$. Köhler himself wrote that this value "is significantly higher than the isotopic signature of the anthropogenic carbon added to the system." That is the single most direct ice-core Keeling intercept measurement in the literature, by one of the standard-literature authors, independently confirming Cohler-Soon's number. Hannon does not mention it in either post.
4. **The arithmetic self-sabotage.** In Post 2, Hannon argues for a natural signature of −11‰ and claims this gives $f \approx 0.13$ rather than $f \approx 0.0$. The IPCC claim she defends is $f \approx 1$. Her own best case leaves the IPCC claim falsified by a factor of nearly 8. She does not mention this.

The cumulative effect across both posts is a cloud of "sciency" wording built on these four failures.

Beyond the itemized rebuttal of each post, three deeper results (end of this document) show that when Hannon's own techniques are applied properly to the real data, they *falsify* the attribution she defends rather than supporting it:

- **Breakpoint analysis done properly.** On the full 1006-2025 record, the real structural break is at 1997-1998, not 1965. The shift is +0.98‰ TOWARD the ocean value, AWAY from the fossil value, during the highest-emissions decades.
- **Recent-window $\delta^{13}C_I$ trend.** Pointwise $\delta^{13}C_I$ on the Francey 1999 dataset shows a trend of +2.94‰ per decade over 1977-1996, against a Hurst-Kolmogorov 95% envelope of [−1.26, +1.24]‰/decade, with $p < 10^{-4}$. The sign is OPPOSITE to the fossil-dominance prediction. Hannon

runs her cumulative regression on a dataset that cuts off three years before the strongest falsification.

- **The mass-balance identity.** The identity cannot support any causal claim. Worked out on the 2010-2019 decadal numbers from IPCC AR6 Figure 5.12, manmade carbon was a NET SINK of about −1.0 GtC/yr while non-manmade sources were a net source of +6.1 GtC/yr. Non-manmade sources drove the entire 5.1 GtC/yr atmospheric growth.

## What each party actually said about the sclerosponge

The central strawman in Hannon's second post is her claim that Cohler-Soon infer an atmospheric Keeling intercept from sclerosponge measurements. The exact wording in each source is as follows.

### Cohler and Soon (2026), Section 4.2 "Five-century proxy extension"

> *"The $\delta^{13}C$ of surface-water dissolved inorganic carbon has been reconstructed from a Caribbean sclerosponge (Ceratoporella nicholsoni) with sub-annual growth banding, extending back to approximately 1500 AD. Sclerosponge skeletal aragonite records DIC $\delta^{13}C$ with a small and stable fractionation offset. The reconstructed surface-water DIC $\delta^{13}C$ translates to an atmospheric $\delta^{13}C_I$ that hovers near $\delta^{13}C_I \approx -13‰$ across the full five-century record, with no significant drift."*

Cohler and Soon (2026) explicitly states that the sclerosponge records surface-water DIC, not atmospheric $\delta^{13}C$. The same paragraph of Section 4.2 explicitly names the fractionation offset and explicitly uses the word "translates to." It does not perform a Keeling regression on sclerosponge data, because a Keeling regression is a mathematical operation on paired atmospheric $\delta^{13}C$ and 1/CO2 data, not on DIC data.

### Koutsoyiannis (2024), Section 2

> *"These were retrieved by digitisation of Figure 4 of Böhm et al., a study that used carbon isotope records of four Caribbean coralline sponges to derive $\delta^{13}C$ records, combined with an ice core record. Here, the atmospheric data of [CO2] and $\delta^{13}C$ from that study are used. These were derived from Antarctic ice core, firn air inclusions and air measurements, while the atmospheric $\delta^{13}C$ record was scaled for its preindustrial mean and minimum values to fit the shallow water sponge record."*

Koutsoyiannis explicitly states that the atmospheric data he digitized from Böhm's Figure 4 came from Antarctic ice core, firn air inclusions, and direct air measurements. Not from the sponges. He explicitly states that the atmospheric $\delta^{13}C$ series was scaled only for visual comparison with the sponge data. His Keeling regression of −13.3‰ (his Figure 14) is a regression on real atmospheric data.

### Böhm et al. (2002), Figure 4 caption

> *"Comparison between Caribbean shallow (blue solid line) and deeper water (red dotted line) sponge $\delta^{13}C$ records (inner left y axis), $\delta^{13}C$ of atmospheric CO2 (green crosses, outer left y axis) and atmospheric pCO2 (circles)... [Etheridge et al., 1996; Francey et al., 1999] and air measurements [Keeling and Whorf, 2001]. The atmospheric $\delta^{13}C$ record is scaled for its preindustrial mean and minimum values to fit the shallow water sponge record."*

Böhm's own Figure 4 plots sponge DIC $\delta^{13}C$ on one axis and real atmospheric $\delta^{13}C$ (from Etheridge 1996, Francey 1999, Keeling-Whorf 2001) on another. The atmospheric axis scaling is a visual overlay. Böhm himself documents both the sponge data and the atmospheric data from independent measurements.

### What Hannon claims Cohler and Soon (2026) says (Post 2)

> *"The paper provides no demonstrated Keeling regression of the sclerosponge data showing how it establishes an atmospheric intercept of −13‰, except for a reference to Böhm et al. (2002)."*
>
> *"The sponge skeletons record the $\delta^{13}C$ of dissolved inorganic carbon (DIC) in seawater. They do not directly measure atmospheric $\delta^{13}C$ and they do not provide corresponding atmospheric CO2 concentrations."*

### Alignment

A Keeling regression is a mathematical operation on paired atmospheric $\delta^{13}C$ and 1/CO2 data. It is not performed on sclerosponge measurements by Cohler-Soon, by Koutsoyiannis, or by Böhm, because sclerosponge measurements are a different data type (surface-water DIC). The atmospheric Keeling regression is performed on real atmospheric data (ice core, firn air, air measurements). The sclerosponge provides the surface-water DIC record that, through the standard gas-DIC fractionation, independently corroborates the five-century stability of the atmospheric signature.

Hannon attacks a position none of the three source papers takes. That is a strawman by the standard definition.

## Post 1 (Nov 2, 2026): "The Keeling Plot and Carbon Isotope Interpretation of Atmospheric CO2"

### Opening paragraph

**Lie 1.1.** "Cohler and Soon's paper... places extraordinary weight on the carbon isotope Keeling intercept." The paper derives its conclusion from three IPCC-endorsed inputs: mass conservation, the standard $\delta^{13}C$ values used throughout the carbon-cycle literature for over six decades, and the primary Scripps and sclerosponge datasets. The intercept is the output. Not a weight.

**Lie 1.2.** "...predominantly natural." Scope shift. The paper's quantitative result is specific: the fossil fraction f of the net atmospheric addition is near zero. It explicitly distinguishes this from the gross fossil inflow (≈4.3%) and the resident fossil stock (≈4.3%). Hannon's phrase blurs the three distinct quantities.

**Lie 1.3.** "...an interpretation of the Keeling relationship that is considerably more straightforward than the actual carbon cycle." The Keeling relation is not an interpretation. It is an algebraic identity derived from mass conservation. All carbon-cycle processes are already summed into the net flux F_net. Its isotopic signature is read off directly.

### "The Keeling intercept is not a definitive source-identification tool"

**Lie 1.4.** "is considered a diagnostic for examining the isotopic composition of $CO_2$ additions to the atmosphere"

FABRICATION. Hannon reframes the Keeling relation as a soft "diagnostic" so it can later be called unreliable. The Keeling relation is a mathematical identity derived from mass conservation: its intercept EQUALS the flux-weighted mean $\delta^{13}C$ of the net atmospheric input. Not an approximate diagnostic tool.

**Lie 1.5.** "The Keeling plot... is particularly useful over relatively short periods." No such restriction exists. The derivation requires only the small-ratio approximation $R \approx 0.011 \ll 1$, which holds at all times in the record. Invented limitation.

**Lie 1.6.** "...extending this two-reservoir interpretation over decades or centuries introduces a fundamental complication: the ocean-atmosphere exchange of CO2 becomes increasingly important." Ocean-atmosphere exchange is already included in the net flux F_net whose signature the intercept measures. Presented as an omitted complication. Not omitted.

**Lie 1.7.** Köhler 2006 inversion. Hannon writes that Köhler et al. (2006) "demonstrated that the isotopic signature inferred from a Keeling plot can differ substantially from the actual isotopic signature of the carbon source." Köhler's actual measurement on Law Dome LD ANT data gave $y_0 = -13.1‰$ with $r^2 = 96\%$. Köhler wrote that this value "is significantly higher than the isotopic signature of the anthropogenic carbon added to the system." That is the Cohler-Soon result in Köhler's own words. Hannon cites Köhler as a refutation of what Köhler measured.

**Lie 1.8.** "They further showed that distinguishing individual carbon-cycle processes using the Keeling approach alone is not straightforward."

FABRICATION. IMPLIES that Cohler and Soon (2026) attempt to distinguish individual sub-processes (photosynthesis, respiration, outgassing, uptake) via Keeling alone. TRUTH: The paper uses a TWO-COMPONENT mass partition between fossil and non-fossil-fuel signatures. It makes no claims about sub-processes. A legitimate Köhler point about sub-process attribution is redirected to attack a claim the paper never made.

**Lie 1.9.** "This is a critical limitation that Cohler and Soon do not adequately resolve."

FABRICATION. IMPLIES a real limitation exists in the paper's method. TRUTH: The "limitation" is Hannon's fabrication, chained to 1.4-1.8. There is nothing for the paper to resolve because the paper did not do what Hannon imputes to it. Assertion in the voice of settled fact.

**Lie 1.10.** "There is also a mathematical issue."

UNSUPPORTED ASSERTION. Bare declaration that an issue exists. No derivation given. The following sentence (1.11) does not supply one either.

**Lie 1.11.** "Their derivation assumes a constant isotopic signature of the net atmospheric carbon addition over the integration interval." False. The paper's integrated form assumes constancy as a hypothesis the data test. The three-way convergence across five centuries (Scripps, sclerosponge, Law Dome) and the $r^2 = 0.96$ of the Law Dome Keeling regression ARE the test. Hannon treats a hypothesis the data confirmed across five centuries as if it were untested.

## "The assertion of a historically invariant intercept is questionable"

**Lie 1.12.** "The assertion of a historically invariant intercept is questionable"

FABRICATION (section header). IMPLIES the paper asserts invariance as an axiom. TRUTH: Cohler and Soon (2026) DEMONSTRATE invariance as an observational RESULT from three independent proxies across five centuries. Reframing a measurement as an "assertion" prepares the reader to treat Hannon's windowing artifact as a counter-measurement.

**Lie 1.13.** "However, an examination of the original Francey et al. (1999) Law Dome atmospheric $CO_2$ and $\delta^{13}C$ record raises questions about this conclusion."

FABRICATION. IMPLIES that Francey et al. themselves raise questions about Cohler and Soon's conclusion. TRUTH: Francey 1999 is an INPUT to the Koutsoyiannis analysis the paper cites. Francey's data is CONSISTENT with the paper. Hannon's own full-record Keeling fit on the same Francey data (admitted in Post 2) gives −12.4 to −13.1‰ - inside the paper's ±0.2‰ band. The only "question" the data raise is one Hannon's windowing scheme manufactures.

**Lie 1.14.** "My cumulative Keeling regression analysis shows considerable variation." Statistical fraud. Early cumulative windows cover spans where atmospheric CO2 barely moved. The 1/C variable has essentially no range in those windows. Fitting a straight line through a tight cluster of 1/C values and extrapolating to the y-axis gives an unstable intercept. Her own $R^2$ (right-axis line in her chart) is near zero in the early windows and rises only as the window widens. The intercepts she displays for the 1860s-1900s are statistical noise.

**Lie 1.15.** "The intercept becomes progressively more negative, eventually reaching approximately −13.1‰." The "progressive" trend is her windowing method stabilizing toward the real answer. The −13‰ at the end is the five-century stable value the paper and Koutsoyiannis document.

**Lie 1.17.** "Nevertheless, the observed variation deserves investigation before historical constancy is treated as an established fact."

FABRICATION. IMPLIES that Cohler and Soon (2026) treat constancy as an "established fact" that she now calls into question. TRUTH: The paper establishes constancy by three independent measurements. The "observed variation" is Hannon's own regression artifact. She uses "established fact" to insinuate an axiom; what she displays is a known statistical property of her chosen method, not a defect in the measurement.

**Lie 1.18.** "Francey et al. (1999) themselves documented substantial decadal-to-century-scale variability in atmospheric $\delta^{13}C$, including a stepped decline between 1860 and 1960." Variable confusion. $\delta^{13}C$ of the reservoir and $\delta^{13}C_I$ of the net addition are different quantities. Francey documents variability in the reservoir composition (as the ice-core record shows and as atmospheric mixing produces). The Keeling relation separates reservoir composition from the signature of the net addition. The intercept is the latter.

## "Carbon isotopes should not be considered in isolation"

**Lie 1.20.** "The continuing decline in atmospheric carbon-13 abundance is consistent with the addition of isotopically light carbon from fossil fuels. This is the well-established Suess effect." The direction of $\delta^{13}C$ change tells you only that the net addition is lighter than the current reservoir. It does not tell you how much lighter. Non-fossil-fuel sources are also light: terrestrial respiration ≈ −25‰, C3 biomass burning ≈ −27‰, methane oxidation ≈ −47‰. The Keeling intercept, which measures the actual signature of the net addition, is −13‰. Fossil is at −28‰. The 15‰ gap is the quantitative issue the "Suess effect" framing hides.

**Lie 1.21.** "The long-term decline in atmospheric O2 provides an independent line of evidence for the combustion of fossil fuels." Scope shift. O2 decline shows fossil fuels are being burned. Nobody disputes that. It does not show the atmospheric CO2 rise is fossil-driven. The Cohler-Soon EHB critique paper (Section 5) walks through this: Salby and Harde (2021) construct a system where the biosphere is a net CO2 sink and net O2 source (matching the O2 constraint) and the ocean is a net CO2 source. Every O2 constraint is satisfied with non-fossil-fuel-dominant attribution.

**Lie 1.22.** "Carbon mass balance: Cumulative fossil-fuel and land-use emissions can be compared with the measured increase in atmospheric carbon and the independently estimated uptake." The mass-balance identity $F_anth - dM_a/dt = F_out - F_nat$ is true in every physically admissible state because it is a rearrangement of mass conservation. A statement true in every state carries zero evidential weight for any causal proposition. Nature-dominated models satisfy it exactly.

**Lie 1.23.** "Ocean and terrestrial sinks: These reservoirs are absorbing substantial quantities of anthropogenic carbon." Begging the question. CO2 molecules are chemically identical regardless of origin. Sinks remove CO2 by chemical identity, not by provenance. Calling what they remove "anthropogenic carbon" assumes the fossil-dominant attribution the Keeling result falsifies.

**Lie 1.24.** "Their uptake and isotope fractionation must be incorporated into any meaningful interpretation of atmospheric isotope changes."

FABRICATION. IMPLIES that Cohler and Soon (2026) fail to incorporate uptake and isotope fractionation. TRUTH: The paper EXPLICITLY uses the standard gas-DIC fractionation of ~9‰ at surface ocean temperatures (Mook 1986; Zhang et al. 1995) and the standard $\delta^{13}C$ value for ocean-outgassed $CO_2$ near −8‰. Fractionation is a load-bearing input, not an omitted one.

**Lie 1.25.** "The Global Carbon Budget provides a particularly useful independent accounting framework." The GCB numbers are a bookkeeping arrangement of the mass-balance identity. The ratio 230/530 ≈ 44% is the cumulative version of the annual rate ratio, not the physical retention fraction. Under proper reservoir routing with $\tau_res \approx 3.8$ yr, the actual retention fraction is approximately 6%, not 44%.

**Lie 1.26.** "attempting to establish source attribution from a single extrapolated isotope intercept"

FABRICATION. IMPLIES Cohler and Soon (2026) use a SINGLE EXTRAPOLATED intercept. TRUTH: The paper attributes from THREE independent intercepts (Scripps modern, sclerosponge via ocean coupling, Köhler's Law Dome Keeling regression) combined with mass conservation and standard $\delta^{13}C$ values. "Single" and "extrapolated" are both false characterizations.

**Lie 1.27.** Citing "(Engelbeen, Hannon, and Burton, 2024)" as authority. Hannon cites her own co-authored CO2 Coalition document in defense of her Substack critique of a paper that explicitly critiques that document. The companion EHB critique by Cohler and Soon addresses every one of EHB's nine lines of evidence.

## Conclusion

**Lie 1.28.** "The Keeling relationship is a valuable tool, but its intercept should not be elevated into a definitive test of anthropogenic $CO_2$ attribution."

FABRICATION. IMPLIES Cohler and Soon (2026) "elevate" the intercept into a "definitive test" it does not deserve. TRUTH: The paper uses the intercept as ONE OF THREE inputs that, taken together, bound the fossil fraction algebraically. "Should not be elevated" presupposes that the paper elevated it. The paper did not; Hannon invented the elevation and then objects to it.

**Lie 1.29.** "A mathematical regression does not, by itself, establish the physical origin of atmospheric carbon." The regression is not alone. Three IPCC-endorsed inputs (mass conservation, standard $\delta^{13}C$ values, primary datasets) force the conclusion. To reject the conclusion requires rejecting one of these. Hannon rejects none of them.

**Lie 1.30.** "Nor does an apparently stable intercept demonstrate that fossil-fuel emissions have made no significant contribution." Equivocation on "contribution." The paper distinguishes three fossil-related fractions. Only the fossil fraction of the net addition is near zero. The other two are approximately 4.3% each.

**Lie 1.31.** "The central question is not whether a particular Keeling regression produces an intercept near −13‰. It is whether that intercept can reliably distinguish the competing sources and sinks operating simultaneously within a dynamic, interconnected carbon cycle."

FABRICATION. IMPLIES Cohler and Soon's (2026) "central question" is sub-process attribution. TRUTH: The paper's central question is the fossil vs non-fossil-fuel partition of the net addition - a two-component mass partition with standard $\delta^{13}C$ values. It does not attempt sub-process attribution. Hannon redefines the paper's question to one it did not ask, so that she can declare it unanswered.

**Lie 1.33.** "The claim that carbon isotope evidence mathematically falsifies fossil-fuel attribution is far stronger than their analysis warrants." Assertion in the voice of settled fact. Hannon has identified no mathematical flaw, rejected none of the three inputs, refuted no proxy, and engaged with none of the three-way convergence.

## Post 2 (Nov 4, 2026): "Sclerosponge Records Cannot Establish a Keeling Intercept"

### Opening paragraphs

**Lie 2.1.** "Cohler and Soon rely on two important premises." Neither item is a premise. The modern intercept near −13‰ is a regression result from paired $\delta^{13}C$ and CO2 data at four Scripps stations. Its historical stability is an observational conclusion from three independent proxies. Hannon reframes measurement results as assumptions.

**Lie 2.2.** "However, examination of the underlying data raises serious questions about both premises."

UNSUPPORTED ASSERTION. Bare declaration. At this point in the post nothing has been shown; "raises serious questions" is asserted in the voice of settled fact before any examination is presented.

**Lie 2.3.** "The modern instrumental record does produce an intercept near −13‰, but this does not establish that the value has remained constant over centuries."

FABRICATION. IMPLIES Cohler and Soon (2026) use the modern four-decade Scripps record to claim century-scale constancy. TRUTH: The paper uses the sclerosponge-coupled ocean record AND Köhler's direct Law Dome ice-core Keeling regression for the century-scale claim, not the modern Scripps record. The Scripps record establishes the modern intercept; the proxies establish stability. Hannon imputes a method the paper did not use.

**Lie 2.4.** "The Caribbean sclerosponge measurements cannot produce an atmospheric Keeling intercept." A Keeling regression is a defined mathematical operation on paired atmospheric $\delta^{13}C$ and 1/CO2 data. No one performs it on sclerosponge measurements, because sclerosponge data is surface-water DIC, a different data type. The atmospheric Keeling regression that produces the −13‰ intercept is Koutsoyiannis's, computed on real atmospheric data (ice core, firn air, air measurements). The sclerosponge separately corroborates five-century stability through the standard gas-DIC fractionation. Straw target.

**Lie 2.5.** "The Law Dome ice-core records show that the inferred Keeling intercept varies depending on the historical period examined." Same cumulative-regression fraud as Post 1. Her own full-record regressions on the same data yield −12.4‰ to −13.1‰, within the paper's ±0.2‰ uncertainty.

### "The Modern Instrumental Record has an Intercept of −13‰"

**Lie 2.6.** "The intercepts actually vary slightly from −12.9‰ at the South Pole to −13.3‰ at La Jolla." Those exact numbers are in the paper (Section 4.1). The ±0.2‰ uncertainty IS the site-to-site variation. She quotes the paper's own figures as if they contradicted it.

**Lie 2.7.** "They cannot establish that the same intercept prevailed in the past... Therefore, proxy data needs to be investigated." The paper does exactly this with three independent proxies: sclerosponge, Köhler's Law Dome ice-core Keeling regression, and Francey/Rubino ice-core series via Koutsoyiannis. Hannon frames the paper's own method as a gap in it.

### "Caribbean Sclerosponge Record Does Not Establish a Keeling Intercept"

**Lie 2.8.** "The Caribbean Sclerosponge record is a fundamental problem with Cohler and Soon's argument." The sclerosponge is corroboration, not foundation. The instrumental Scripps data alone restrict f to [0, 0.26]. The sclerosponge tightens that to $f \approx 0$ through the stability result. Even deleted entirely, the paper's result still falsifies the IPCC claim of $f \approx 1$ by at least a factor of four.

**Lie 2.9.** "The paper provides no demonstrated Keeling regression of the sclerosponge data showing how it establishes an atmospheric intercept of −13‰." No one performs a Keeling regression on sclerosponge data, because a Keeling regression is a mathematical operation on paired atmospheric $\delta^{13}C$ and 1/CO2 data.

Section 4.2 explicitly states the sclerosponge records surface-water DIC and that the DIC record "translates to" atmospheric $\delta^{13}C_I$ through the small, stable gas-DIC fractionation offset. The atmospheric regression itself is Koutsoyiannis's, computed on real atmospheric data.

**Lie 2.10.** "The paper invokes the claimed five-century, approximately −13‰ sclerosponge result at least 10 additional times." Citation count is not an argument. The result is cited many times because it is used many times. The question is whether it holds. Köhler's direct Law Dome Keeling regression ($y_0 = -13.1$‰, $r^2 = 96\%$) independently confirms it.

## "What Böhm et al. (2002) actually measured"

**Lie 2.11.** "What Böhm et al. (2002) actually measured"

FABRICATION (subsection header). IMPLIES Cohler and Soon (2026) misrepresented what Böhm measured. TRUTH: Section 4.2 of the paper describes Böhm accurately: "Sclerosponge skeletal aragonite records DIC $\delta^{13}C$ with a small and stable fractionation offset." Nothing to correct. The header exists to prime the reader that a correction is coming when none is.

**Lie 2.12.** "The sponge skeletons record the $\delta^{13}C$ of dissolved inorganic carbon (DIC) in seawater. They do not directly measure atmospheric $\delta^{13}C$..." True. Also irrelevant. Surface-water DIC and atmospheric CO2 are coupled by gas exchange. Ocean DIC at +1.5‰ plus the ≈9‰ equilibrium gas-DIC fractionation gives an ocean-outgassed signature near −8‰. This is standard isotopic oceanography (Mook 1986; Zhang et al. 1995). The sclerosponge record of surface-water DIC therefore constrains the ocean side of the atmospheric isotope budget.

**Lie 2.13.** "This distinction is critical."

FABRICATION. IMPLIES a critical distinction exists between what Böhm did and what Cohler and Soon (2026) claimed Böhm did. TRUTH: No such distinction exists. The paper describes Böhm accurately. "Critical" signals a load-bearing contrast that is not there.

**Lie 2.14.** "In section 4.2, Cohler and Soon acknowledge the following..."

FABRICATION (loaded framing). IMPLIES the paper "acknowledges" a weakness it was forced to concede. TRUTH: Cohler and Soon (2026) STATES the translation step as part of its argument. "Acknowledge" is a loaded verb that converts a straightforward statement of method into a reluctant admission. The accurate verb is "states" or "explains."

**Lie 2.15.** "There is no translation to an atmospheric Keeling intercept in the Böhm et al. (2002) article derived from their sponge data." Correct that Böhm did not do a Keeling plot. Irrelevant to the paper. Böhm published the sponge DIC record AND plotted it alongside real atmospheric $\delta^{13}C$ data (Etheridge 1996, Francey 1999, Keeling-Whorf 2001) in his Figure 4. Koutsoyiannis performed the atmospheric Keeling regression on the atmospheric data. Cohler-Soon cites both.

## "Where did the purported five-century Keeling intercept come from?"

**Lie 2.16.** "Where did the purported five-century Keeling intercept come from?"

FABRICATION (subsection header). IMPLIES the five-century Keeling intercept is "purported" - i.e., alleged without basis. TRUTH: The five-century intercept is published in Koutsoyiannis (2024) Figure 14, from a regression on real atmospheric data (Antarctic ice core, firn air, Keeling-Whorf direct air measurements). "Purported" is doing the work of insinuating fraud or fabrication where there is a published, replicable analysis.

**Lie 2.17.** "Koutsoyiannis explains that he digitized Figure 4 from Böhm et al. (2002)... The atmospheric measurements are from Antarctic ice-core, firn-air, and direct atmospheric observations from the work of

Etheridge et al. (1996) and Francey et al. (1999)." This sentence is Hannon's own admission that the atmospheric data is real atmospheric data, not sponge data. It defeats her own thesis. If the atmospheric Keeling intercept comes from real atmospheric data, then "the sclerosponge cannot produce a Keeling intercept" is attacking a position no one holds.

**Lie 2.18.** "The reported preindustrial means illustrate the distinction: approximately −6.37‰ for atmospheric $\delta^{13}C$ versus +4.95‰ for the sponge record." Her own numbers demonstrate the ocean coupling she denies. An 11‰ offset between atmospheric $\delta^{13}C$ and sponge DIC $\delta^{13}C$ is what gas-DIC equilibrium fractionation predicts, plus the small vital-effect offset typical of sponge skeletons. She presents the signature of the physical coupling as if it were evidence of no coupling.

**Lie 2.19.** "This scaling does not transform the sponge measurements into atmospheric measurements." True. Attacking a non-operation: no one in the literature treats sponge measurements as transformed atmospheric measurements. The sponge record and the atmospheric record are two independent measurements that co-vary because they are physically linked by gas exchange.

**Lie 2.20.** "Consequently, the historical atmospheric data that produced the −13 Keeling intercept reported by Koutsoyiannis (2024) is simply the ice core atmospheric data shown in Figure 1." This is Hannon admitting the −13‰ intercept comes from real atmospheric data. Her own Figure 1 shows the Koutsoyiannis "Bohm" series and the Law Dome Francey 1999 series lying on top of each other. That is visual agreement between two independent measurements, not a problem.

## "A Changing Keeling Intercept in Ice Core Records" / "Full-record regressions"

**Lie 2.21.** "The minor differences in Koutsoyiannis' data from the raw ice core data are probably due to digitization from a paper figure."

UNSUPPORTED ASSERTION. "Probably due to digitization" is speculation presented in the voice of settled fact. It also undercuts her own thesis: two independent digitizations of Böhm's Figure 4 giving nearly identical atmospheric data CONFIRMS the data's robustness, which she spins as a minor artifact.

**Lie 2.22.** "A Changing Keeling Intercept in Ice Core Records"

FABRICATION (section header). IMPLIES the intercept is changing - stated as fact in the header before any evidence is presented. TRUTH: The intercept is not changing. Properly measured on the full 1006-2025 record, every CGAA-containing window sits in [−12.79, −13.77]‰. The "change" is a windowing artifact that the header presents as a result.

**Lie 2.23.** "The Antarctic Law Dome ice cores provide a much more direct way to investigate historical atmospheric $CO_2$ and $\delta^{13}C$."

FABRICATION. IMPLIES Cohler and Soon's (2026) method was indirect or circuitous and this new analysis is a more direct alternative. TRUTH: The paper uses the SAME Law Dome data (via Francey 1999 and Köhler 2006), plus Scripps instrumental data, plus the sclerosponge-coupled ocean record. Hannon's "much more direct way" is the paper's own primary data source.

**Lie 2.24.** "Full-record regressions produce intercepts ranging from approximately −12.4‰ to −13.1‰, with relatively high coefficients of determination." Buried headline. Her own full-record Keeling regressions reproduce the paper's −13.2 ± 0.2‰ result. She then switches to broken sub-windows and presents the broken result as if it discredited her own working one.

**Lie 2.25.** "However, a single regression across the entire record from about 1000 to 1993 AD can conceal changes in the underlying relationship." A single regression on the full record is the correct test for whether the relation is single or multi-regime. Her own $r^2$ values are high. The test passed. She frames a successful test as if it had hidden something.

## "Structural break around 1965"

**Lie 2.26.** "Structural break around 1965"

FABRICATION (subsection header). IMPLIES a 1965 structural break is established as fact. TRUTH: No structural break exists at 1965 in the atmospheric signature. The "break" coincides with the Law Dome data-source transition (ice bubbles to firn air to direct atmospheric sampling). Run properly on the full 1006-2025 record, the real break is at 1997-1998, and it shifts the intercept TOWARD the ocean value.

**Lie 2.27.** "The intercept pre-1965 is −11.0‰ and post-1965 is −13.0‰." Her Figure 2 shows pre-1965 red points clustering between 1/CO2 ≈ 0.00352 and 0.00357. That is a 1/CO2 range of 0.00005. The slope and intercept of a straight line through such a tight cluster are highly correlated: tiny slope differences give large extrapolated-intercept differences. The post-1965 window spans 1/CO2 ≈ 0.00287 to 0.00310, roughly five times wider. The pre-1965 −11‰ is a broken extrapolation. The post-1965 −13‰ is a real Keeling regression. The "2‰ shift" is the difference between a broken regression and a real one. Not a physical change.

**Lie 2.28.** "The analysis identified the break in both the Francey and Rubino datasets between 1965-1967, sup-F = 36.4, bootstrap p = 0.03." The sup-F test detects where residual structure changes. The window she identified coincides with the Law Dome data-source transition (ice-core bubbles to firn-air to direct atmospheric sampling). Different measurement systems. Different sampling artifacts. Finding a "break" at the data-source transition is finding a data-source transition.

**Lie 2.29.** "These results challenge the assumption that one constant Keeling intercept adequately describes the entire historical record." The intercept is not an assumption. It is the quantity being measured. Her own full-record fits already answered the question.

## "Cumulative Keeling intercept becomes increasingly lighter"

**Lie 2.30.** "Cumulative Keeling intercept becomes increasingly lighter"

FABRICATION (subsection header). IMPLIES a cumulative intercept change is an established finding. TRUTH: The "change" is a windowing artifact, not a change in the intercept. The header packages the artifact as the finding.

**Lie 2.31.** The cumulative Keeling regression itself. The same statistical fraud as Post 1, detailed in its own section below.

**Lie 2.32.** "The resulting pattern is revealing as shown in Figure 3."

UNSUPPORTED ASSERTION. "Is revealing" is editorializing. What Figure 3 actually shows (per her own $R^2$ curve) is that early narrow-window intercepts have $R^2$ near zero. That is not a revelation; it is a textbook regression artifact.

**Lie 2.33.** "From late 1800 through 1960, the cumulative intercept oscillates around −11‰ to −11.5‰. After 1960, it begins a pronounced decline, eventually reaching approximately −13.1‰ by 1993." The "pronounced decline" is the regression stabilizing as the 1/C range widens. The stable end value is her own confirmation of the paper's result.

**Lie 2.34.** "The results are shown only for the portion of the record where there are sufficient observations and acceptable regression fit."

SELF-SABOTAGE. Quiet methodological concession. She admits that part of the series had insufficient data and unacceptable fit, and she cut it. But she presents the remaining "cleaned-up" artifact as if it were evidence, when the early-window intercepts near −11‰ that anchor her whole conclusion are themselves on the edge of the rejected unreliable-fit territory.

**Lie 2.35.** "Lastly, there is another relevant result in Koutsoyiannis (2024) that deserves attention. His Figure 15 presents a rolling Keeling intercept calculated from his 'Böhm' atmospheric ice core series using his fixed $CO_2$ difference technique. The results show considerable temporal variability such as periods of increasing trends, relative stability and decreasing trends."

FABRICATION. IMPLIES Koutsoyiannis's own Figure 15 shows variability that supports Hannon's thesis and contradicts Cohler and Soon (2026). TRUTH: Koutsoyiannis's conclusion from his analysis is "net isotopic signature... no change since the Little Ice Age" (title of his paper). Short-window rolling fits produce noise; longer windows stabilize at −13.2‰. Hannon selectively cites the short-window variability while omitting Koutsoyiannis's conclusion that the series is stable overall. See the companion Cohler-Soon b_K extension paper, which runs this test properly on the extended 1006-2025 record and gets b_K = −13.42 ± 0.07‰, $r^2 = 0.988$.

**Lie 2.36.** "This progressive change and variability suggest that the historical relationship is more complicated than a single fixed intercept." Her own full-record fits already contradicted this. She manufactures variability by her choice of windowing scheme, then cites the manufactured variability.

### Conclusions

**Lie 2.37.** "Cohler and Soon's claim that the Keeling intercept stayed near −13% for five centuries rests on a sclerosponge record that contains no atmospheric Keeling regression." Omits Köhler 2006 entirely. Köhler directly computed the Keeling regression on Law Dome LD ANT ice-core data and got $y_0 = -13.1$‰, $r^2 = 96\%$. That is an atmospheric Keeling regression. She does not cite it anywhere in her post.

**Lie 2.38.** "That record is not stable and shows an intercept of about −11‰ before the 1960s, shifting to about −13‰ afterward." Her own full-record Keeling regression on the Law Dome data produced −12.4 to −13.1‰. She computed this herself, reported it, then dismissed it in favor of her broken sub-windows.

**Lie 2.39.** "Their natural baseline of −13‰ therefore has no direct support." Köhler 2006 is direct support. The Scripps four-site modern record is direct support. Her own full-record Law Dome fits are direct support. "No direct support" requires deleting all three.

**Lie 2.40.** "Using it in their own Eq. 12 gives $f \approx 0.13$ rather than $f \approx 0.0$." Arithmetic check granted. Using $\delta^{13}C_N = -11$‰ with $\delta^{13}C_I = -13.2$‰ and $\delta^{13}C_F = -28$‰ gives $f = (-13.2 + 11) / (-28 + 11) \approx 0.129$. The IPCC claim she defends is $f \approx 1$. Her best case leaves that claim falsified by a factor of about 7.7. She does not mention this.

**Lie 2.41.** "This raises an important question: What caused this change?" The change is a windowing artifact. There is no physical change to explain.

**Lie 2.42.** "That question requires examining the relationship between atmospheric isotope changes, natural carbon-cycle exchanges, and increasing anthropogenic emissions."

FABRICATION. IMPLIES the relationship between isotope changes, non-fossil-fuel exchanges, and fossil emissions has not been examined. TRUTH: Cohler and Soon (2026) and the companion EHB critique paper examine this relationship in detail: the three-way isotopic convergence, the oxygen partition, the mass balance, the CERES energy imbalance route, and the bomb-14C constraint. Closing invitation to "further examine" a question that has been examined and answered.

## The cumulative Keeling regression: why it is statistically broken

Hannon's central original contribution across both posts is a procedure she calls "cumulative Keeling regression." She fixes the earliest observation, then adds progressively younger observations, recalculating the Keeling intercept at each step. She reports intercepts near −11‰ in the earlier windows, drifting to −13‰ as the window widens, and presents this as evidence of a time-varying atmospheric signature.

The procedure is statistically broken for the following reasons.

## Keeling plots require range in 1/C

A Keeling plot regresses $\delta^{13}C$ against 1/[CO2]. The intercept is the y-axis value at 1/[CO2] = 0, i.e., infinite CO2. In practice the data never sit at 1/C = 0. The intercept is an extrapolation from the data to the y-axis. The quality of that extrapolation depends on the range the data cover in 1/C. Wide range gives you a well-constrained slope and therefore a well-constrained intercept. Narrow range gives you a slope whose uncertainty is large compared to its value, and an extrapolated intercept whose uncertainty swamps the fitted value.

## The early cumulative windows cover almost no 1/C range

Atmospheric CO2 barely moved during the preindustrial period. For centuries CO2 held near 280 ppm. $1/280 \approx 0.00357$. A window from 1000 AD to 1860 AD spans a 1/C range of perhaps 0.00005. A window to 1900 AD, still very narrow. The extrapolation distance to the y-axis is roughly 0.00357. So the window spans about 1.4% of the distance to the extrapolation target.

Fitting a straight line through a tight cluster at $1/C \approx 0.00355$ and extrapolating to 1/C = 0 is mathematically equivalent to taking the slope of a line through an unstable short segment and projecting it across more than seventy times its length. Small slope differences translate into large intercept differences. The fitted intercept carries essentially no information about the true y-intercept value.

## Her own chart proves the broken fits are noise

Hannon plots $R^2$ on a second y-axis in her Figure 3 cumulative-intercept charts. In the early windows (1860 to around 1920 in her Francey chart, 1900 to around 1935 in her Rubino chart) the $R^2$ values are low. Near zero in the earliest windows. These $R^2$ values indicate that the regression explains almost none of the variance. That is the textbook sign of an under-powered regression on a narrow 1/C range.

As the window widens past the preindustrial-only era and atmospheric CO2 begins to rise, 1/C range grows. $R^2$ climbs. The intercept, previously unstable, begins to stabilize toward its true value. By the time $R^2$ reaches 0.9 or higher, the intercept has settled near −13‰.

The "drift" from −11‰ to −13‰ that Hannon displays is the regression stabilizing. Not a physical shift in the atmospheric signature.

## The structural break at 1965 is the same artifact

Hannon also reports a sup-F structural break test identifying a break at 1965-1967. Two effects produce this result. First, the Law Dome data source transitions around that window: pre-1965 data are ice-core bubbles with gas-age distribution smoothing; post-1965 data bring in firn-air measurements and transition to direct atmospheric sampling. Different measurement systems, different sampling artifacts, different noise characteristics. A structural break test sensitive to residual structure will fire at a measurement-system transition.

Second, the pre-1965 and post-1965 windows have very different 1/C ranges. Pre-1965: 1/CO2 spans roughly 0.00352 to 0.00357, a range of 0.00005. Post-1965: 1/CO2 spans 0.00287 to 0.00310, a range of 0.00023, roughly five times wider. The pre-1965 fit is a broken extrapolation. The post-1965 fit is a real Keeling regression. The intercepts differ for the same reason that the cumulative fits drift.

## Her own full-record regressions confirm Cohler and Soon (2026)

Hannon reports that full-record regressions on both the Francey and Rubino datasets "produce intercepts ranging from approximately −12.4‰ to −13.1‰, with relatively high coefficients of determination." Her

own full-record numbers reproduce Cohler and Soon (2026)'s −13.2 ± 0.2‰ result to within that paper's stated uncertainty. She buries this finding and switches to the broken sub-windows.

### Direct ice-core confirmation she omits

Köhler et al. (2006) performed a Keeling regression directly on Law Dome LD ANT ice-core data and obtained $y_0 = -13.1‰$ with $r^2 = 96\%$. That is a real, published, peer-reviewed atmospheric Keeling regression on Law Dome data, by Köhler himself, giving the same answer as the Scripps modern record, the sclerosponge-coupled ocean record, and Hannon's own full-record Law Dome fits. The $r^2 = 96\%$ says the straight-line relation holds across the entire Law Dome record. No sub-period structure challenges it when the 1/C range is adequate.

Hannon's windowing scheme does not challenge the physical quantity. It demonstrates that narrow-window Keeling regressions on preindustrial-only data cannot constrain the intercept, which is a statistical fact about regression and not a new finding.

## The Köhler 2006 omission

Köhler, Fischer, Schmitt and Munhoven (2006), *On the application and interpretation of Keeling plots in paleo climate research*, performed a Keeling regression directly on Law Dome LD ANT ice-core atmospheric $\delta^{13}C$ and CO2 data. They reported an intercept of $y_0 = -13.1‰$ with $r^2 = 96\%$.

Köhler himself wrote in that paper that the measured value "is significantly higher than the isotopic signature of the anthropogenic carbon added to the system."

This is the single most direct ice-core atmospheric Keeling regression in the peer-reviewed literature. It was performed by Köhler, one of the standard-literature authors in paleoclimate carbon-cycle modeling. It uses the same Law Dome record Hannon re-analyzes in her Post 2. It gives the same −13‰ result that Cohler and Soon (2026), Koutsoyiannis, the Scripps instrumental data, and Hannon's own full-record fits produce.

Köhler 2006 does not appear in either Hannon post. Not in the references. Not in the text.

The Cohler-Soon paper cites Köhler 2006 as the Law Dome ice-core confirmation (Section 4.3): "Kohler et al. (2006) analyzed the Law Dome ice-core record and reported a Keeling regression of $\delta^{13}C$ against 1/C that yields an intercept of approximately −13‰ with $r^2 = 0.96$."

The Cohler-Soon EHB critique paper (Section 3.4) further documents that Köhler himself, in subsequent papers, has admitted failures of every principal architectural element of his 2006 BICYCLE paleoclimate model, including:

- Köhler, Fischer and Schmitt (2010): "Northern Hemisphere temperature is probably incorrect" in the 2006 model.
- Köhler and Munhoven (2020): wholesale replacement of the 2006 carbonate compensation mechanism.
- Köhler et al. (2022): "prescribing CO2 violates mass conservation of carbon" (the exact practice used for the 2006 $r^2 = 0.89$ result).
- Köhler, Skinner and Adolphi (2024): "Opposite to data CO2 falls during Heinrich stadials" (wrong sign of the AMOC response).

The 2006 Keeling intercept measurement, however, stands as direct data. It is not a model output. It is a regression on observed ice-core atmospheric $\delta^{13}C$ and CO2 data. And Köhler himself noted that the result contradicts fossil-dominant attribution.

A critique of the five-century "sclerosponge stability" result that does not mention the direct ice-core Keeling intercept measurement of the same value, by the same author Hannon elsewhere relies on in the paleoclimate literature, is a critique with a load-bearing omission at the center.

## Pattern summary

Across both Hannon posts, five recurring devices do the work. None is a scientific argument. All five fail against the primary sources.

### 1. Strawman

She refutes operations no one performs. A Keeling regression is a mathematical operation on paired atmospheric $\delta^{13}C$ and 1/CO2 data. It is not done on sclerosponge measurements by Cohler-Soon, by Koutsoyiannis, or by Böhm, because sclerosponge data is a different data type (surface-water DIC). The atmospheric Keeling regression uses real atmospheric data (ice core, firn air, air measurements), computed by Koutsoyiannis. The sclerosponge supplies the independent surface-water DIC record that corroborates through ocean-atmosphere isotopic coupling. Hannon spends two sections of Post 2 attacking an operation nobody performs.

### 2. Unsupported assertion in the voice of settled fact

Phrases like "far stronger than their analysis warrants," "fundamental complication," "critical limitation," "fundamental problem," "not stable," "no direct support" appear with no derivation and no citation at the point of assertion. She states conclusions as if established.

### 3. Non-sequitur

Correct observations are followed by attribution conclusions that do not follow from them. $\delta^{13}C$ is declining, so fossil fuels caused it. O2 is declining, so fossil fuels caused the CO2 rise. GCB numbers balance, so attribution follows. Every one of these premises is compatible with non-fossil-fuel-dominant attribution.

### 4. Omission as refutation

Köhler 2006 is the strongest single piece of evidence against her sclerosponge argument. She omits it from both posts. The companion Cohler-Soon EHB critique paper addresses every one of EHB's nine lines of evidence in detail. She does not mention it. The three-way convergence across Scripps, sclerosponge, and Law Dome is what tightens the fossil-fraction bound to zero. She does not engage with it.

### 5. Statistical artifact as evidence

The cumulative Keeling regression is a windowing scheme that produces artifactual intercept drift from narrow 1/C windows. The structural-break test fires at the Law Dome data-source transition. Her own full-record Keeling regressions on the same data reproduce Cohler and Soon (2026)'s $-13 \pm 0.2$‰ result. She buries that agreement and presents the artifact as the real finding.

## Rhetorical character

The pattern across both posts is "sciency" wording layered over mathematical and bibliographic failures. The individual sentences sound measured. The internal logic does not survive contact with the primary sources.

Each post ends with a "this raises an important question" or "far stronger than warranted" style rhetorical flourish. These are not conclusions earned by the arguments. They are closing phrases substituted for arguments that were not made.

### Scoreboard

Even on her own best case, her Post 2 Equation 12 arithmetic gives f ≈ 0.13 using her −11‰ preferred baseline. The IPCC-EHB claim she defends is f ≈ 1. Her best-case assault on Cohler and Soon (2026) leaves that claim falsified by a factor of roughly 7.7. She does not mention this. Cohler and Soon (2026)'s central thesis, that fossil-fuel combustion contributes a negligible fraction of the net atmospheric CO2 addition, survives her own strongest argument against it.

## Hannon's breakpoint analysis is backwards (and falsifies fossil dominance when done properly)

Hannon reports a "structural break at 1965-1967" from her sup-F scan on Rubino 2019 and Francey 1999 ice-core data. The result is an artifact of two choices: (a) the dataset she used terminates in 1993 or 1996 and misses the key modern-era data, and (b) she places the break where the data source transitions (ice core bubbles to firn air measurements), not where the atmospheric signature actually shifts.

When the same test is run properly on the full 1006-2025 record (combining the Francey Law Dome ice-core data, the Cape Grim Air Archive flasks from Francey's Table 1, and the CSIRO Cape Grim CGO5011 in situ monthly record through 2025), the real structural break appears at 1997-1998, not 1965. And the direction of the shift falsifies fossil dominance by direct measurement.

### The actual full-record Keeling regression

On 101 annual pairs spanning 1006-2025, the full-record Keeling intercept is $b_K = -13.42 \pm 0.07$‰ (OLS 1σ), with $r^2 = 0.988$. The HAC-robust standard error (accounting for residual serial correlation) is ±0.10‰. The intercept is statistically indistinguishable from the Koutsoyiannis four-site Scripps result of −13.2 ± 0.2‰, and 15‰ removed from the standard fossil value of −28‰.

Every CGAA-containing window tested, from 1750-2025 through 1850-2025 through 1978-2025, falls in the band [−12.79, −13.77]‰. Nowhere does −11‰ appear. Hannon's −11‰ preindustrial baseline is the narrow-window extrapolation artifact. The real baseline, on real data properly regressed, is between −13.4‰ and −13.8‰ across every tested span.

### The real structural break: 1997-1998

The Chow / Andrews sup-F scan on the full record peaks at 1971. That peak sits at the ice-core-to-modern platform transition and is a data-source artifact, as noted in Section 5 above and confirmed by the recursive sub-scan.

The recursive sub-scan restricted to the modern segment (1971-2025, N = 47) peaks at 1997, with 1998 and 1999 as the next two candidates. This locates the real structural break in the atmospheric signature at 1997-1998.

### The shift is TOWARD the ocean value, AWAY from fossil

Pre-break fit, 1978-1998 (N = 16): $b_K = -13.77 \pm 0.33$‰, $r^2 = 0.960$.

Post-break fit, 1998-2025 (N = 28): $b_K = -12.79 \pm 0.05$‰, $r^2 = 0.997$.

The shift is $\Delta b_K = +0.98$‰. The post-break fit is extraordinarily clean, with $r^2$ of 0.997 and RMSE of 0.011‰ on 28 annual points.

The Welch-style test of the intercept difference, HAC-robust against the residual serial correlation, gives $t = 2.66$ with $p = 0.018$. The shift is statistically significant.

The direction of the shift is critical. The intercept moves from −13.77‰ toward −12.79‰. That is a move TOWARD the standard ocean-outgassing value of −8‰ and AWAY from the fossil value of −28‰. During the highest-emissions decades.

Under fossil dominance, b_K must move toward −28‰. It moves the other way. The direct measurement contradicts the prediction.

### The 1997-1998 coincidence

The 1997-1998 break date coincides with three independent large-scale forcing events: the onset of the CERES-measured planetary albedo decline, the AMO flip to positive phase, and the PDO flip to negative phase, together with the completion of post-Pinatubo stratospheric aerosol recovery. Each is a natural, non-fossil-fuel forcing. The isotopic shift and the forcing shift coincide in time.

### Scoreboard

Hannon's conclusion: "the inferred intercept is sensitive to the period being examined and the underlying isotopic relationships have changed."

The real finding, on the properly extended record, is sharper: the Keeling intercept is stationary at −13.4 ± 0.4‰ across a thousand years, with one discrete regime shift at 1997-1998 that moves the signature TOWARD the ocean value during the highest-emissions decades. That is the opposite of fossil-dominance evidence. That is fossil-dominance falsification by direct measurement.

## The recent-window $\delta^{13}C_I$ trend is positive and significant, directly falsifying fossil dominance

Hannon's whole framing treats the Keeling intercept as a static object whose constancy versus variability is the question. There is a sharper test: compute the pointwise $\delta^{13}C_I$ over the highest-emissions window and look at the trend.

Under fossil dominance, the trend must be negative. This follows from the mass partition equation by direct differentiation: if f (the fossil fraction of the net addition) grows with the fossil-fuel emission rate, and fossil-fuel emissions accelerated through the 1977-1996 window, then $d(\delta^{13}C_I)/dt$ must be negative over that window. The first term in the derivative (proportional to $\delta^{13}C_F - \delta^{13}C_N$, i.e., −28 minus roughly −13, so about −15) is unambiguously negative times a growing f. The second term ($d\delta^{13}C_N/dt$) is bounded to less than about 0.01‰/decade by the five-century sclerosponge stability of the non-fossil-fuel signature. So the second term cannot change the sign of the first.

Under fossil dominance the recent-window trend in $\delta^{13}C_I$ must therefore be negative.

### The direct measurement

Applying Koutsoyiannis's Equation 13 with the fixed-ΔCO2 technique to the Francey 1999 record (Law Dome ice cores, firn air, and Cape Grim Air Archive) over 1977-1996:

Observed trend: $d(\delta^{13}C_I)/dt$ = **+2.94‰ per decade**.

Hurst-Kolmogorov 95% envelope (10,000 Monte Carlo replicates at the observed Hurst exponent of 0.921): [−1.26, +1.24]‰ per decade.

p-value against the H-K null: $\mathbf{p < 10^{-4}}$.

The observed trend is positive in sign, more than twice the upper bound of the noise envelope in that opposite direction, and statistically significant at above 99.99% confidence against a long-range persistent noise null.

### What this means

The observed sign is OPPOSITE to the fossil-dominance prediction. The observed magnitude exceeds the upper bound of the noise envelope by a factor of 2.4 in the opposite direction.

The Hurst-Kolmogorov noise model is intentionally conservative. It allows for long-range persistence in the geophysical series and inflates the noise envelope relative to a white-noise test. A rejection at $p < 10^{-4}$ against the H-K null is a stronger statement than a rejection against white noise.

The fossil-dominance hypothesis is falsified by the direct measurement on the same Francey 1999 dataset that Hannon uses for her cumulative Keeling regression.

### Robustness

Applying the same procedure to the Rubino 2013/2019 revision (which terminates at 1993, three years before the sharpest phase of the upward reversal) gives +0.67‰ per decade ($p = 0.40$). Restoring the three CGAA tank measurements for 1994-1996 to the Rubino ice-and-firn base (same CSIRO GASLAB measurement chain, same CG92 calibration scale, same Cape Grim Air Archive) recovers +1.81‰ per decade ($p < 10^{-4}$).

The sign is positive on every treatment. The result is robust to data-source choice once the endpoint truncation is corrected.

### What Hannon missed

Hannon runs her cumulative Keeling regression on the Rubino dataset which terminates in 1993. That dataset omits the Cape Grim Air Archive tanks and endpoint-truncates three years before the sharpest phase of the recent upward reversal. She is cutting the data off precisely where the fossil-dominance falsification is strongest.

The positive recent-window trend is the direct falsification. Hannon's cumulative-regression framing hides what the pointwise $\delta^{13}C_I$ series does when actually computed.

## The mass-balance "evidence" is not evidence at all

Hannon's Lie 1.13, 1.14, and 1.15 (Post 1) and the entire EHB (2024) report she defends rely on the same technique: present the atmospheric carbon mass-balance identity as evidence for anthropogenic attribution. The technique is mathematically invalid at the most elementary level. An accounting identity cannot serve as evidence for any claim about cause. The error has been named independently across economics, medicine, demography, fluid dynamics, and double-entry bookkeeping. It is not peculiar to climate.

### What the identity actually says

Start with atmospheric mass conservation: $dM_a/dt = F_in - F_out$. One quantity is directly measured: $M_a$, from Mauna Loa and the NOAA network. Its time derivative follows by differentiation. $F_in$ and $F_out$ are not directly measured anywhere on Earth. They are definitional quantities whose values are fixed by the bookkeeping scheme itself.

The scheme then splits the input into a non-fossil-fuel part and a fossil-fuel part: $F_in \equiv F_nat + F_anth$. $F_anth$ is estimated from bottom-up inventories of fossil-fuel combustion, cement, and land-use change. $F_nat$ is not measured anywhere; it enters only through the split as the complement.

Substituting and rearranging gives the commonly cited form:

**$F_anth - dM_a/dt = F_out - F_nat.$**

This adds no empirical content to the two inputs it was derived from. It is a bookkeeping rearrangement. It holds in every situation in which the mass-conservation postulate and the fossil-vs-non-fossil-fuel split hold.

## Why it carries zero evidential weight

A statement that is true in every physically admissible state cannot distinguish one state from another. A statement that cannot distinguish states cannot change the probability of any hypothesis about which state we are in. In Bayesian terms, the likelihood ratio $P(E \mid C) / P(E \mid \neg C)$ equals $1/1 = 1$ for any causal proposition C, because the identity E is realized in every state regardless of C. The posterior equals the prior. Learning the identity shifts the credence of any causal claim by exactly zero.

In plain terms: the identity is true whether fossil fuels are driving the atmospheric rise or non-fossil-fuel sources are. It cannot pick between them. Presenting it as if it could is a logical error, not a physical argument.

## Stacking identities does not help

EHB (2024) and Hannon's post both offer multiple framings of the same rearrangement: the airborne fraction, the sink uptake fraction, the land-ocean partition, cumulative emissions versus cumulative atmospheric growth. All of these are rearrangements of the one identity. Each individually carries zero evidential weight. Their intersection (what Bayesian updating would combine) is again the whole state space, so the aggregate carries zero evidential weight too. The "multiple lines of evidence" framing is empty when every line is the same identity rewritten.

## The 2010-2019 decadal numbers

Working through the identity with the IPCC AR6 Figure 5.12 and Global Carbon Budget numbers for 2010-2019 produces a picture that is the OPPOSITE of the IPCC framing.

- Atmospheric reservoir $M_a \approx 850$ GtC.
- Gross input $F_{in} \approx 226.9$ GtC/yr.
- Gross output $F_{out} \approx 221.8$ GtC/yr.
- Net growth $dM_a/dt = F_{in} - F_{out} \approx 5.1$ GtC/yr.
- Total anthropogenic emission $F_{anth} \approx 11.1$ GtC/yr (fossil plus land-use minus cement carbonation).
- Non-fossil-fuel input $F_{nat} \approx 215.8$ GtC/yr.

The atmospheric residence time is $\tau_{res} = M_a / F_{out} \approx 3.8$ yr. Because CO2 molecules are chemically indistinguishable by origin in a well-mixed reservoir, every sub-pool decays at the same rate as the bulk. The convolutional integral of the GCB anthropogenic series back to 1959 under exponential decay at $\tau_{res}$ gives a resident manmade sub-pool of approximately 46.1 GtC. The manmade outflow follows as pool divided by $\tau_{res}$: about 12.1 GtC/yr.

**Manmade outflow (12.1) exceeds manmade inflow ($F_{anth}$ = 11.1). Manmade carbon was a NET SINK of approximately −1.0 GtC/yr over 2010-2019.**

**Non-manmade sources ($F_{nat}$ = 215.8 in, non-manmade outflow 209.7 out) were a NET SOURCE of approximately +6.1 GtC/yr.**

**The entire 5.1 GtC/yr atmospheric growth came from non-manmade sources. Manmade activity slightly offset it.**

The identity is satisfied: 11.1 − 5.1 = 221.8 − 215.8 = 6.0 GtC/yr. The same identity is satisfied under the IPCC framing where nature is a net sink and manmade activity drives the rise. The two framings cannot be distinguished by the identity. That is precisely the point of the formal result above.

### What this means for Hannon's citations

Hannon's Lie 1.13 (carbon mass balance), Lie 1.14 (ocean and terrestrial sinks absorbing "anthropogenic" carbon), and Lie 1.15 (Global Carbon Budget cumulative framing) all invoke the same empty rearrangement. Her citation of EHB (2024) inherits EHB's central technique, which is exactly the technique the formal result above forbids.

The mass-balance identity is where the IPCC and EHB claims rest. Removing it removes the base. The isotopic and radiocarbon falsifications (sections above) then operate on actual observations rather than on bookkeeping rearrangements, and those actual observations give $f \approx 0$.

## Conclusion

Two Substack posts by Renee Hannon attack Cohler and Soon (2026). Across them, seventy-five passages tagged in the companion forensic review misrepresent what Cohler and Soon (2026) computes, what the cited primary sources report, and what the published atmospheric dataset shows. The pattern is not accidental and not confined to one writer.

The technique has a documented history. It originated in the tobacco industry's 1960s defence of its products against established health findings (the Brown & Williamson "Doubt is our product" memo). It was refined across decades of industry-funded science denial through acid rain, ozone, pesticides, and climate attribution. It is now standard practice in state-level information operations. The variant on display in these two posts is the mathematical one: converting defined mathematical operations and their results into "claims" that can be disputed without finding any arithmetic error. The academic names are *manufactured doubt* and *epistemological downgrading*.

What Cohler and Soon (2026) computes is a mathematical operation on published atmospheric datasets. The operation returns a number with its uncertainty: −13.2 ± 0.2‰ from the Scripps four-site data via Koutsoyiannis; −13.1‰ with $r^2 = 96\%$ from the Law Dome ice-core data via Köhler et al. (2006); approximately −13‰ across five centuries from the sclerosponge surface-water DIC record, through the standard gas-DIC fractionation; and −12.4 to −13.1‰ from Hannon's own full-record Keeling regression on the same Law Dome data. All four agree. Under the mass partition equation with the standard $\delta^{13}C$ values for fossil-fuel combustion (−28‰) and ocean-outgassed CO2 (−8‰), this intercept forces the fossil fraction of the net atmospheric addition to $f \approx 0$. The IPCC claim is $f \approx 1$. The two numbers are not reconcilable.

Hannon does not find an arithmetic error in Cohler and Soon (2026). She does not reject any of the three input categories. She does not refute any of the three independent proxies. She does not cite the Köhler 2006 measurement at all. Her original contribution, the cumulative Keeling regression on narrow windows, produces artifactual intercept drift from a known statistical defect: extrapolating a straight line to the y-axis from a tight cluster of 1/C values gives an intercept whose uncertainty swamps the fitted value. Her own full-record fits reproduce Cohler and Soon (2026)'s result and she buries that agreement. Her own best-case arithmetic ($f \approx 0.13$ using her preferred −11‰ baseline) leaves the IPCC claim of $f \approx 1$ falsified by a factor of nearly 8, and she does not mention this.

The posts do not challenge the physics. They convert the physics into opinions. The response to them cannot be another restatement of the physics. The response has to be what this Catalog does: name the technique, show how each instance operates, and refuse to let the conversion proceed.

Mathematics is not opinion. A Keeling regression computed on a published dataset is not a claim. A regression result with its uncertainty is not something one can simply disagree with.